

Redesigning a Train Ticket Machine Interface: A User-Centered Approach

Bernard Kučina
bk@student.uni-lj.si
Fakulteta za računalništvo in
informatiko
Ljubljana, Slovenia

Jan Gračner
jg@student.uni-lj.si
Fakulteta za računalništvo in
informatiko
Ljubljana, Slovenia

Strahinja Dorđević
sAŠ@student.uni-lj.si
Fakulteta za računalništvo in
informatiko
Ljubljana, Slovenia

Marko Horvat
mh@student.uni-lj.si
Fakulteta za računalništvo in
informatiko
Ljubljana, Slovenia

Gašper Suhadolnik
gs1874@student.uni-lj.si
Fakulteta za računalništvo in
informatiko
Ljubljana, Slovenia

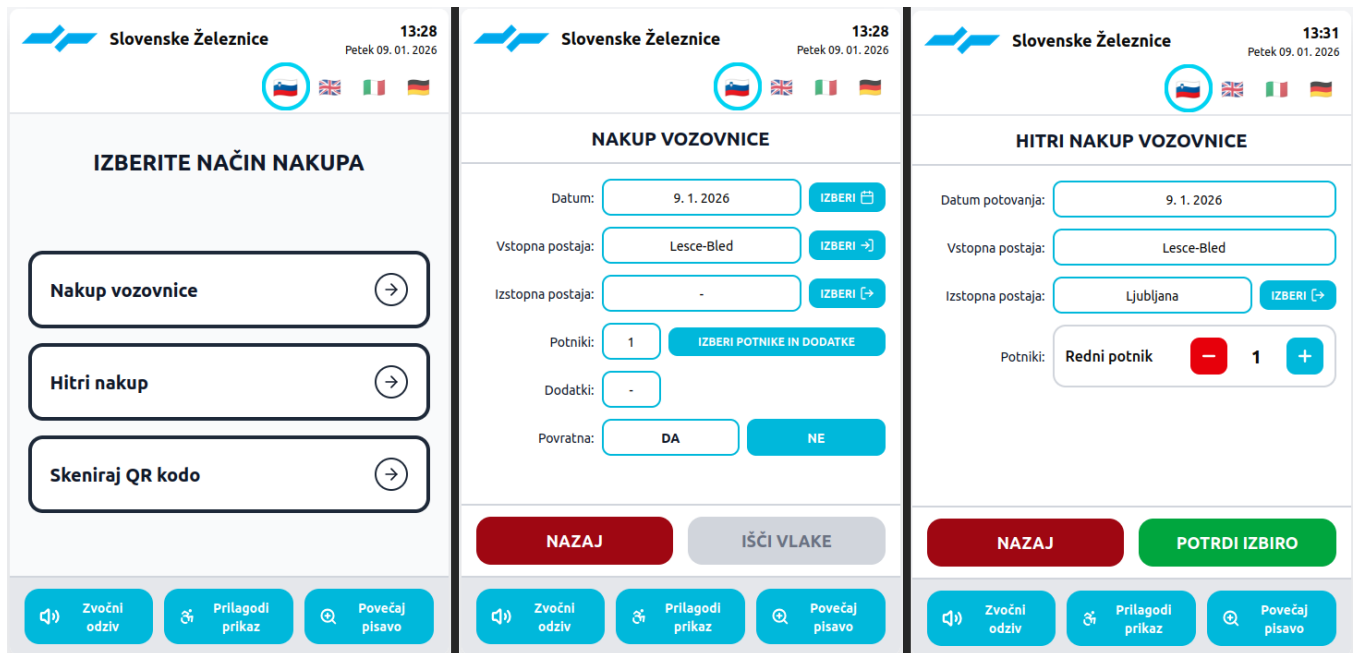


Figure 1: Redesigned user interface of a train ticket machine.

Abstract

Train ticket machines are a common point of interaction for a wide range of users, yet they are often perceived as difficult to use. This project explores the redesign of a human-computer interface for a train ticket machine using a user-centered design process. We

first conducted an online questionnaire (N=20) to identify common usability issues and user expectations. Based on these findings and developer insights, we created an initial prototype in Figma. The prototype was evaluated through interviews and informal usability testing, which informed several iterative design improvements. The final prototype addresses key usability concerns such as clarity, efficiency, and ease of use. This paper presents the design process, findings, and lessons learned, while detailed data and materials are provided in the project portfolio.

Keywords

ticket machine, human-computer interface, redesign, prototype, portfolio, testing

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1 Introduction

Train ticket machines are widely used in public transportation systems and must accommodate users with diverse backgrounds, technical skills, and time constraints. Poorly designed interfaces can lead to confusion, delays, and frustration, negatively affecting the overall travel experience. Despite their importance, many ticket machines still suffer from usability issues such as complex navigation, unclear terminology, and inefficient workflows.

The goal of this project was to design an improved human-computer interface for a train ticket machine that better supports user needs and expectations. We adopted a user-centered design approach, combining user research, prototyping, and iterative evaluation. The main research questions guiding this project were:

- What usability issues do users experience with current train ticket machines?
- Which interface changes do users perceive as most beneficial for improving their experience?

2 Methodology

The project followed an iterative design process consisting of four main phases: user research, initial prototyping, user evaluation, and redesign.

2.1 Questionnaire Study

We designed an online questionnaire to gather user opinions and experiences with existing train ticket machines. The questionnaire was distributed to approximately 20 participants. It included questions about common difficulties, desired features, and overall satisfaction. The collected data was analyzed to identify recurring themes and key usability problems. Detailed questionnaire data and analysis scripts are available in the project portfolio.

2.2 Initial Prototype

Based on the questionnaire results and our own analysis as developers, we designed an initial interface prototype using Figma. The prototype focused on simplifying the ticket purchasing flow, improving visual clarity, and reducing cognitive load.

2.3 User Interviews and Feedback

The initial prototype was presented to several users, who were asked to explore the interface and provide feedback through semi-structured interviews. Participants commented on usability, clarity, and perceived improvements compared to existing machines.

2.4 Iterative Redesign

Using the feedback from interviews, we refined the prototype to address identified issues. Changes included layout adjustments, clearer labeling, and improved navigation flow. The final prototype reflects these iterative improvements.

3 Related work

TODO

4 Results

This section presents the findings from the questionnaire study and the user evaluation of the prototype. Quantitative summaries, anonymized responses, and supporting materials are provided in the project portfolio.

4.1 Questionnaire Results

A total of 20 participants completed the questionnaire regarding their experiences with existing train ticket machines. Most respondents were students and working adults who reported frequent use of ticket machines.

4.1.1 Overall experience. Participants generally rated their experience with current ticket machines as poor. Notably, dissatisfaction was also expressed by frequent users, suggesting that usability issues persist even after repeated exposure to the interface.

4.1.2 Interaction complexity. One of the most frequently reported issues was the excessive number of steps required to complete a ticket purchase. Participants described the process as slow and unnecessarily complex, particularly in situations where time pressure was present.

4.1.3 Interface clarity and input problems. Many respondents mentioned unclear instructions and confusing terminology. Small button sizes were highlighted as problematic for touch interaction. In addition, several participants complained about the on-screen keyboard, describing it as inaccurate and difficult to use when entering destinations.

4.1.4 Missing functionality. Participants also identified limitations in available features. The interface was reported to support only two languages (Slovenian and English), which some users considered insufficient in an international travel context. Additionally, users noted the absence of an option to add a bicycle fee to the final ticket price, despite certain trains allowing bicycles on board and requiring an additional charge.

(Detailed questionnaire responses and distributions are included in the project portfolio.)

4.2 Interview and Prototype Evaluation Results

The initial prototype was evaluated by approximately ten participants across four informal testing sessions. Users interacted with the prototype and provided feedback through semi-structured interviews.

4.2.1 Quick buy feature feedback. Participants generally responded positively to the introduction of a “quick buy” page intended to streamline the purchasing process. However, multiple users reported that the feature lacked flexibility. In particular, they noted the inability to quickly add additional passengers or optional services such as a bicycle fee.

4.2.2 Information availability. Several participants expressed a desire to see more contextual information directly on the quick buy page. Most notably, users wanted to view information about the

next available train they could board, which they felt would support faster and more informed decision-making.

4.2.3 Guidance and transparency. Some users requested clearer instructions for using the ticket machine, especially for first-time or infrequent users. Participants also expressed interest in being able to access terms and conditions related to ticket purchases directly within the interface.

4.2.4 Visual design feedback. Overall, the visual design of the prototype was well received. Minor issues were reported regarding spacing, button emphasis, and visual hierarchy. These comments informed small but meaningful adjustments in the subsequent iteration of the prototype.

(Graphs summarizing questionnaire results and screenshots of the prototype iterations are included as figures in this paper.)

5 Discussion

This section discusses the implications of the results presented above and reflects on how the findings address the research questions of this project.

5.1 Addressing identified usability issues

The questionnaire results highlighted several persistent usability problems in existing train ticket machines, including interaction complexity, unclear instructions, small touch targets, and limited functionality. These findings directly informed the design goals of the prototype. Feedback from the prototype evaluation suggests that many of these issues were partially addressed, particularly through a simplified interaction flow and clearer visual structure. Users perceived the redesigned interface as easier to understand and more efficient, supporting the effectiveness of a user-centered and iterative design approach.

5.2 Balancing efficiency and flexibility

The introduction of a "quick buy" feature was intended to reduce the number of steps required for frequent users. While participants responded positively to this concept, feedback revealed that excessive simplification can reduce flexibility. The inability to quickly add passengers or optional services, such as a bicycle fee, highlighted the need to balance efficiency with configurability. This finding emphasizes that shortcuts in public interfaces should still allow access to essential options without forcing users into alternative workflows.

5.3 Information transparency and user confidence

Requests for clearer instructions, access to terms and conditions, and contextual information such as the next available train indicate that users value transparency and reassurance when interacting with public ticketing systems. Providing such information can increase user confidence, particularly for infrequent users or tourists, and may reduce errors and hesitation during the purchasing process.

5.4 Design implications for public ticket machines

The results suggest that effective ticket machine interfaces should prioritize clarity, large and accessible touch targets, multilingual support, and complete coverage of real-world use cases, such as bicycle transport. Iterative testing with real users proved essential for uncovering both major usability problems and smaller design details that significantly affect user experience.

5.5 Limitations

This study has several limitations. The questionnaire sample size was relatively small and consisted mainly of students and working adults, which may limit the generalizability of the findings. The prototype evaluation was informal and conducted with a limited number of participants. Additionally, the evaluation focused on perceived usability rather than objective performance measures such as task completion time or error rates.

Despite these limitations, the findings provide valuable insights into common usability challenges and demonstrate the benefits of iterative, user-centered design for public transportation interfaces.

6 Conclusion and future work

This paper presented a user-centered redesign of a train ticket machine interface aimed at improving usability, clarity, and efficiency for a broad range of users. Through an initial questionnaire study, we identified key usability problems in existing ticket machines, including complex interaction flows, unclear instructions, small touch targets, and missing functionality. These findings guided the design of an initial prototype, which was iteratively refined based on user interviews and feedback.

The results of the prototype evaluation indicate that the redesigned interface was perceived as easier to understand and more efficient than existing systems. In particular, users responded positively to the simplified layout and streamlined purchasing flow. At the same time, the evaluation highlighted the importance of maintaining flexibility and transparency in public interfaces, as demonstrated by feedback regarding optional services, contextual information, and access to instructions and terms.

Future work could extend this project in several directions. First, the prototype could be evaluated through a more formal usability study with a larger and more diverse participant sample, including tourists, elderly users, and users with accessibility needs. Second, future iterations could incorporate additional features such as expanded language support, accessibility options, and tighter integration with real-time train data. Finally, deploying and testing the interface in a real-world setting would allow for the collection of objective performance metrics, such as task completion time and error rates, providing deeper insight into the effectiveness of the proposed design.

Overall, this project demonstrates the value of iterative, user-centered design in improving human-computer interfaces for public transportation systems and highlights key considerations for the design of future ticketing machines.

References